

Lecture Notes: Techniques of Integration

Calculus I

Author: Bakdaulet Sotsial

Astana IT University

Contents

1	Integration by Parts	2
2	Special Cases of Integration	3
3	Integrals of Rational Functions by Partial Fractions	6

Section 1 Integration by Parts

Theorem 1 (Integration by Parts Formula) If $u = f(x)$ and $v = g(x)$ are differentiable functions, then

$$\int u \, dv = uv - \int v \, du.$$

In terms of definite integrals,

$$\int_a^b u \, dv = [uv]_a^b - \int_a^b v \, du.$$

The product rule for differentiation states that

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}.$$

Integrating both sides with respect to x gives

$$uv = \int u \, dv + \int v \, du.$$

Rearranging yields $\int u \, dv = uv - \int v \, du$.

Remark 1 (Choosing u and dv) A useful guideline for choosing u is the acronym LIATE: Logarithmic, Inverse trigonometric, Algebraic, Trigonometric, Exponential. The function that appears earlier in this list should generally be chosen as u , and the remaining part as dv .

Example 1 Evaluate $\int x e^x \, dx$. Let $u = x$ and $dv = e^x \, dx$. Then $du = dx$ and $v = e^x$. Applying the formula,

$$\int x e^x \, dx = x e^x - \int e^x \, dx = x e^x - e^x + C = e^x(x - 1) + C.$$

Example 2 Evaluate $\int \ln x \, dx$. Let $u = \ln x$ and $dv = dx$. Then $du = \frac{1}{x} \, dx$ and $v = x$. Thus,

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

Example 3 Evaluate $\int x \sin x \, dx$. Let $u = x$ and $dv = \sin x \, dx$. Then $du = dx$ and $v = -\cos x$. Thus,

$$\int x \sin x \, dx = -x \cos x - \int (-\cos x) \, dx = -x \cos x + \int \cos x \, dx = -x \cos x + \sin x + C.$$

Example 4 Evaluate $\int x^2 e^x \, dx$. Apply integration by parts twice. First, let $u = x^2$ and $dv = e^x \, dx$. Then $du = 2x \, dx$ and $v = e^x$:

$$\int x^2 e^x \, dx = x^2 e^x - 2 \int x e^x \, dx.$$

From the previous example, $\int x e^x \, dx = e^x(x - 1) + C$. Therefore,

$$\int x^2 e^x \, dx = x^2 e^x - 2e^x(x - 1) + C = e^x(x^2 - 2x + 2) + C.$$

Example 5 Evaluate $\int e^x \sin x \, dx$. Let $I = \int e^x \sin x \, dx$. Apply integration by parts with $u = \sin x$ and $dv = e^x \, dx$. Then $du = \cos x \, dx$ and $v = e^x$:

$$I = e^x \sin x - \int e^x \cos x \, dx.$$

Apply integration by parts again to $\int e^x \cos x \, dx$ with $u = \cos x$ and $dv = e^x \, dx$. Then $du = -\sin x \, dx$ and $v = e^x$:

$$\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I.$$

Substituting back,

$$I = e^x \sin x - (e^x \cos x + I) = e^x \sin x - e^x \cos x - I.$$

Thus, $2I = e^x(\sin x - \cos x)$, so

$$I = \frac{e^x}{2}(\sin x - \cos x) + C.$$

Example 6 Evaluate $\int \arctan x \, dx$. Let $u = \arctan x$ and $dv = dx$. Then $du = \frac{1}{1+x^2} \, dx$ and $v = x$. Thus,

$$\int \arctan x \, dx = x \arctan x - \int \frac{x}{1+x^2} \, dx.$$

For the remaining integral, let $w = 1 + x^2$, so $dw = 2x \, dx$:

$$\int \frac{x}{1+x^2} \, dx = \frac{1}{2} \ln(1+x^2) + C.$$

Therefore,

$$\int \arctan x \, dx = x \arctan x - \frac{1}{2} \ln(1+x^2) + C.$$

Example 7 Evaluate $\int x \ln x \, dx$. Let $u = \ln x$ and $dv = x \, dx$. Then $du = \frac{1}{x} \, dx$ and $v = \frac{x^2}{2}$. Thus,

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C.$$

Section 2 Special Cases of Integration

Theorem 2 (Reduction Formula for $\int \sin^n x \, dx$) For $n \geq 2$,

$$\int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx.$$

Write $\sin^n x = \sin^{n-1} x \sin x$ and integrate by parts with $u = \sin^{n-1} x$ and $dv = \sin x \, dx$. Then $du = (n-1) \sin^{n-2} x \cos x \, dx$ and $v = -\cos x$. Thus,

$$\int \sin^n x \, dx = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx.$$

Using $\cos^2 x = 1 - \sin^2 x$,

$$\int \sin^n x \, dx = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx.$$

Adding $(n-1) \int \sin^n x \, dx$ to both sides gives

$$n \int \sin^n x \, dx = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \, dx.$$

Dividing by n yields the formula.

Theorem 3 (Reduction Formula for $\int \cos^n x \, dx$) For $n \geq 2$,

$$\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx.$$

Similar to the sine reduction formula, using $u = \cos^{n-1} x$ and $dv = \cos x \, dx$.

Example 8 Evaluate $\int \sin^4 x \, dx$. Using the reduction formula with $n = 4$,

$$\int \sin^4 x \, dx = -\frac{1}{4} \sin^3 x \cos x + \frac{3}{4} \int \sin^2 x \, dx.$$

For $\int \sin^2 x \, dx$, use the identity $\sin^2 x = \frac{1-\cos 2x}{2}$.

$$\int \sin^2 x \, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

Therefore,

$$\int \sin^4 x \, dx = -\frac{1}{4} \sin^3 x \cos x + \frac{3}{4} \left(\frac{x}{2} - \frac{\sin 2x}{4} \right) + C = -\frac{1}{4} \sin^3 x \cos x + \frac{3x}{8} - \frac{3 \sin 2x}{16} + C.$$

Theorem 4 (Integrals of the Form $\int \tan^n x \, dx$ and $\int \sec^n x \, dx$) For $n \geq 2$,

$$\int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx.$$

For $n \geq 2$,

$$\int \sec^n x \, dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx.$$

For the tangent formula, write $\tan^n x = \tan^{n-2} x \tan^2 x = \tan^{n-2} x (\sec^2 x - 1)$. Then

$$\int \tan^n x \, dx = \int \tan^{n-2} x \sec^2 x \, dx - \int \tan^{n-2} x \, dx.$$

The first integral is $\frac{\tan^{n-1} x}{n-1}$ by substitution $u = \tan x$. For the secant formula, integrate by parts with $u = \sec^{n-2} x$ and $dv = \sec^2 x \, dx$.

Example 9 Evaluate $\int \tan^3 x \, dx$. Using the formula with $n = 3$,

$$\int \tan^3 x \, dx = \frac{\tan^2 x}{2} - \int \tan x \, dx = \frac{\tan^2 x}{2} + \ln |\cos x| + C.$$

Theorem 5 (Integrals of the Form $\int \sin^m x \cos^n x \, dx$) 1. If m is odd, substitute $u = \cos x$.

2. If n is odd, substitute $u = \sin x$.

3. If both m and n are even, use the half-angle identities

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

Example 10 Evaluate $\int \sin^3 x \cos^2 x \, dx$. Since $m = 3$ is odd, let $u = \cos x$. Then $du = -\sin x \, dx$ and $\sin^2 x = 1 - \cos^2 x = 1 - u^2$. Thus,

$$\int \sin^3 x \cos^2 x \, dx = \int \sin^2 x \cos^2 x \sin x \, dx = \int (1-u^2)u^2(-du) = \int (u^4-u^2) \, du = \frac{u^5}{5} - \frac{u^3}{3} + C = \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$

Example 11 Evaluate $\int \sin^2 x \cos^4 x \, dx$. Both exponents are even. Use $\sin^2 x = \frac{1-\cos 2x}{2}$ and $\cos^2 x = \frac{1+\cos 2x}{2}$:

$$\sin^2 x \cos^4 x = \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 + \cos 2x}{2} \right)^2 = \frac{1}{8} (1 - \cos 2x)(1 + 2 \cos 2x + \cos^2 2x).$$

Expanding,

$$= \frac{1}{8} (1 + 2 \cos 2x + \cos^2 2x - \cos 2x - 2 \cos^2 2x - \cos^3 2x) = \frac{1}{8} (1 + \cos 2x - \cos^2 2x - \cos^3 2x).$$

Using $\cos^2 2x = \frac{1+\cos 4x}{2}$,

$$= \frac{1}{8} \left(1 + \cos 2x - \frac{1 + \cos 4x}{2} - \cos^3 2x \right) = \frac{1}{16} (1 + 2 \cos 2x - \cos 4x - 2 \cos^3 2x).$$

Thus,

$$\int \sin^2 x \cos^4 x \, dx = \frac{1}{16} \left(x + \sin 2x - \frac{\sin 4x}{4} - 2 \int \cos^3 2x \, dx \right).$$

For $\int \cos^3 2x \, dx$, let $u = \sin 2x$, $du = 2 \cos 2x \, dx$, and $\cos^2 2x = 1 - u^2$:

$$\int \cos^3 2x \, dx = \int \cos^2 2x \cos 2x \, dx = \frac{1}{2} \int (1 - u^2) \, du = \frac{1}{2} \left(u - \frac{u^3}{3} \right) = \frac{\sin 2x}{2} - \frac{\sin^3 2x}{6}.$$

Substituting back yields the final answer.

Theorem 6 (Integrals of the Form $\int \sin(mx) \cos(nx) \, dx$) Use the product-to-sum identities:

$$\sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)],$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)],$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)].$$

Example 12 Evaluate $\int \sin 3x \cos 5x \, dx$. Using the identity,

$$\sin 3x \cos 5x = \frac{1}{2} [\sin(3x - 5x) + \sin(3x + 5x)] = \frac{1}{2} [\sin(-2x) + \sin 8x] = \frac{1}{2} [-\sin 2x + \sin 8x].$$

Thus,

$$\int \sin 3x \cos 5x \, dx = \frac{1}{2} \left(\frac{\cos 2x}{2} - \frac{\cos 8x}{8} \right) + C = \frac{\cos 2x}{4} - \frac{\cos 8x}{16} + C.$$

Section 3 Integrals of Rational Functions by Partial Fractions

Definition 1 (Rational Function) A rational function is a function of the form $f(x) = \frac{P(x)}{Q(x)}$, where P and Q are polynomials and $Q(x) \neq 0$.

Definition 2 (Proper and Improper Rational Functions) A rational function $\frac{P(x)}{Q(x)}$ is proper if the degree of P is less than the degree of Q . It is improper if the degree of P is greater than or equal to the degree of Q .

Theorem 7 (Partial Fraction Decomposition) Any proper rational function $\frac{P(x)}{Q(x)}$ can be expressed as a sum of simpler fractions, provided $Q(x)$ is factored completely into linear and irreducible quadratic factors. The form of the decomposition depends on the factors of $Q(x)$:

1. For each distinct linear factor $(ax + b)$, include a term $\frac{A}{ax+b}$.
2. For each repeated linear factor $(ax + b)^k$, include terms $\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_k}{(ax+b)^k}$.
3. For each distinct irreducible quadratic factor $(ax^2 + bx + c)$, include a term $\frac{Ax+B}{ax^2+bx+c}$.
4. For each repeated irreducible quadratic factor $(ax^2 + bx + c)^k$, include terms $\frac{A_1x+B_1}{ax^2+bx+c} + \cdots + \frac{A_kx+B_k}{(ax^2+bx+c)^k}$.

Example 13 Evaluate $\int \frac{1}{x^2 - 1} dx$. Factor the denominator: $x^2 - 1 = (x - 1)(x + 1)$. Write

$$\frac{1}{(x - 1)(x + 1)} = \frac{A}{x - 1} + \frac{B}{x + 1}.$$

Multiply by $(x - 1)(x + 1)$: $1 = A(x + 1) + B(x - 1)$. Set $x = 1$: $1 = 2A \implies A = \frac{1}{2}$. Set $x = -1$: $1 = -2B \implies B = -\frac{1}{2}$. Thus,

$$\int \frac{1}{x^2 - 1} dx = \frac{1}{2} \int \frac{1}{x - 1} dx - \frac{1}{2} \int \frac{1}{x + 1} dx = \frac{1}{2} \ln|x - 1| - \frac{1}{2} \ln|x + 1| + C = \frac{1}{2} \ln \left| \frac{x - 1}{x + 1} \right| + C.$$

Example 14 Evaluate $\int \frac{2x + 3}{x^2 + 3x + 2} dx$. Factor: $x^2 + 3x + 2 = (x + 1)(x + 2)$. Write

$$\frac{2x + 3}{(x + 1)(x + 2)} = \frac{A}{x + 1} + \frac{B}{x + 2}.$$

Multiply: $2x + 3 = A(x + 2) + B(x + 1)$. Set $x = -1$: $1 = A(1) \implies A = 1$. Set $x = -2$: $-1 = B(-1) \implies B = 1$. Thus,

$$\int \frac{2x + 3}{x^2 + 3x + 2} dx = \int \frac{1}{x + 1} dx + \int \frac{1}{x + 2} dx = \ln|x + 1| + \ln|x + 2| + C = \ln|(x + 1)(x + 2)| + C.$$

Example 15 Evaluate $\int \frac{x^2 + 2x - 1}{x^3 - x} dx$. Factor: $x^3 - x = x(x - 1)(x + 1)$. Write

$$\frac{x^2 + 2x - 1}{x(x - 1)(x + 1)} = \frac{A}{x} + \frac{B}{x - 1} + \frac{C}{x + 1}.$$

Multiply: $x^2 + 2x - 1 = A(x - 1)(x + 1) + Bx(x + 1) + Cx(x - 1)$. Set $x = 0$: $-1 = A(-1)(1) \implies A = 1$. Set $x = 1$: $1 + 2 - 1 = 2 = B(1)(2) \implies B = 1$. Set $x = -1$: $1 - 2 - 1 = -2 = C(-1)(-2) \implies C = -1$. Thus,

$$\int \frac{x^2 + 2x - 1}{x^3 - x} dx = \int \frac{1}{x} dx + \int \frac{1}{x - 1} dx - \int \frac{1}{x + 1} dx = \ln|x| + \ln|x - 1| - \ln|x + 1| + C = \ln \left| \frac{x(x - 1)}{x + 1} \right| + C.$$

Example 16 Evaluate $\int \frac{x^2 + 1}{x(x^2 + 1)} dx$ (note: this simplifies before decomposition). Actually, $\frac{x^2 + 1}{x(x^2 + 1)} = \frac{1}{x}$ for $x \neq 0$. Thus,

$$\int \frac{x^2 + 1}{x(x^2 + 1)} dx = \int \frac{1}{x} dx = \ln|x| + C.$$

Example 17 Evaluate $\int \frac{1}{x^2(x + 1)} dx$. The factor x^2 is a repeated linear factor. Write

$$\frac{1}{x^2(x + 1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x + 1}.$$

Multiply: $1 = Ax(x + 1) + B(x + 1) + Cx^2$. Set $x = 0$: $1 = B(1) \implies B = 1$. Set $x = -1$: $1 = C(1) \implies C = 1$. Expand and compare coefficients: $1 = A(x^2 + x) + (x + 1) + x^2 = (A + 1)x^2 + (A + 1)x + 1$. Thus $A + 1 = 0 \implies A = -1$. Therefore,

$$\int \frac{1}{x^2(x + 1)} dx = -\int \frac{1}{x} dx + \int \frac{1}{x^2} dx + \int \frac{1}{x + 1} dx = -\ln|x| - \frac{1}{x} + \ln|x + 1| + C = \ln\left|\frac{x + 1}{x}\right| - \frac{1}{x} + C.$$

Example 18 Evaluate $\int \frac{2x + 1}{x^2 + 2x + 5} dx$. The denominator $x^2 + 2x + 5$ is irreducible (discriminant $4 - 20 = -16 < 0$). Complete the square: $x^2 + 2x + 5 = (x + 1)^2 + 4$. Write the numerator as $2x + 1 = 2(x + 1) - 1$. Thus,

$$\int \frac{2x + 1}{x^2 + 2x + 5} dx = \int \frac{2(x + 1) - 1}{(x + 1)^2 + 4} dx = \int \frac{2(x + 1)}{(x + 1)^2 + 4} dx - \int \frac{1}{(x + 1)^2 + 4} dx.$$

For the first integral, let $u = (x + 1)^2 + 4$, $du = 2(x + 1) dx$:

$$\int \frac{2(x + 1)}{(x + 1)^2 + 4} dx = \ln((x + 1)^2 + 4) + C_1.$$

For the second, let $v = \frac{x + 1}{2}$, $dv = \frac{1}{2} dx$:

$$\int \frac{1}{(x + 1)^2 + 4} dx = \frac{1}{2} \arctan\left(\frac{x + 1}{2}\right) + C_2.$$

Therefore,

$$\int \frac{2x + 1}{x^2 + 2x + 5} dx = \ln(x^2 + 2x + 5) - \frac{1}{2} \arctan\left(\frac{x + 1}{2}\right) + C.$$

Example 19 Evaluate $\int \frac{x^3 + x}{x - 1} dx$. The rational function is improper. Perform polynomial long division:

$$\frac{x^3 + x}{x - 1} = x^2 + x + 2 + \frac{2}{x - 1}.$$

Thus,

$$\int \frac{x^3 + x}{x - 1} dx = \int \left(x^2 + x + 2 + \frac{2}{x - 1}\right) dx = \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln|x - 1| + C.$$

Theorem 8 (General Method for Integrating Rational Functions) 1. If the rational function is improper, perform polynomial long division.

2. Factor the denominator completely into linear and irreducible quadratic factors.

3. Decompose the proper rational function into partial fractions.

4. Integrate each partial fraction term using standard formulas.

Example 20 Evaluate $\int \frac{4x^2 + 3x + 2}{x^3 + x^2} dx$. Factor: $x^3 + x^2 = x^2(x + 1)$. Write

$$\frac{4x^2 + 3x + 2}{x^2(x + 1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x + 1}.$$

Multiply: $4x^2 + 3x + 2 = Ax(x + 1) + B(x + 1) + Cx^2$. Set $x = 0$: $2 = B \implies B = 2$. Set $x = -1$: $4 - 3 + 2 = 3 = C(1) \implies C = 3$. Expand: $4x^2 + 3x + 2 = A(x^2 + x) + 2x + 2 + 3x^2 = (A + 3)x^2 + (A + 2)x + 2$. Comparing coefficients: $A + 3 = 4 \implies A = 1$. Thus,

$$\int \frac{4x^2 + 3x + 2}{x^3 + x^2} dx = \int \frac{1}{x} dx + 2 \int \frac{1}{x^2} dx + 3 \int \frac{1}{x + 1} dx = \ln|x| - \frac{2}{x} + 3 \ln|x + 1| + C.$$

Example 21 Evaluate $\int \frac{x^2 - 2x - 3}{(x - 1)(x^2 + 1)} dx$. Write

$$\frac{x^2 - 2x - 3}{(x - 1)(x^2 + 1)} = \frac{A}{x - 1} + \frac{Bx + C}{x^2 + 1}.$$

Multiply: $x^2 - 2x - 3 = A(x^2 + 1) + (Bx + C)(x - 1)$. Set $x = 1$: $1 - 2 - 3 = -4 = A(2) \implies A = -2$. Expand: $x^2 - 2x - 3 = -2x^2 - 2 + Bx^2 - Bx + Cx - C = (B - 2)x^2 + (C - B)x + (-2 - C)$. Comparing coefficients: $B - 2 = 1 \implies B = 3$. $C - B = -2 \implies C - 3 = -2 \implies C = 1$. Check constant: $-2 - C = -3 \implies -2 - 1 = -3$, correct. Thus,

$$\int \frac{x^2 - 2x - 3}{(x - 1)(x^2 + 1)} dx = -2 \int \frac{1}{x - 1} dx + \int \frac{3x + 1}{x^2 + 1} dx.$$

For the second integral,

$$\int \frac{3x + 1}{x^2 + 1} dx = \frac{3}{2} \int \frac{2x}{x^2 + 1} dx + \int \frac{1}{x^2 + 1} dx = \frac{3}{2} \ln(x^2 + 1) + \arctan x + C.$$

Therefore,

$$\int \frac{x^2 - 2x - 3}{(x - 1)(x^2 + 1)} dx = -2 \ln|x - 1| + \frac{3}{2} \ln(x^2 + 1) + \arctan x + C.$$